Unit 1 - Sem 4 - 22MAT230

Mathematics for Computing 4

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https://github.com/mfcpjt/MFC4_22MAT230

```
clearvars
ready = true;
PUBLISH = ready;
```

PCA

is called unsupervised learning because it finds patterns and structure in data without using known output labels or target variables, focusing solely on the input feature's inherent variance to reduce dimensions, unlike supervised methods that need correct answers to learn from. It identifies principal components (new features) that capture the most data variation, making it great for simplifying data (like reducing noise) or preprocessing for later supervised tasks, but its core process remains label-free.

- **Finds Intrinsic Structure:** It discovers hidden patterns by finding directions (principal components) of maximum data spread (variance) within the data itself.
- **Feature Extraction/Reduction:** Its goal is to create fewer, new features (components) that summarize the original data, not to predict an outcome.

Analogy:

Imagine sorting a pile of mixed fruits (apples, oranges, bananas) without knowing what each fruit is called (labels). PCA is like someone who notices the different colors, shapes, and sizes, and then groups them based

on these inherent qualities (e.g., "round & red," "long & yellow") to simplify the pile, even if they don't know the fruit's name.

What PCA Learns (The Core Concepts):

Pattern Discovery: It finds inherent patterns (variance) and structure, making it great for simplifying data for visualization or improving other algorithms.

1. **Data Variation:** PCA identifies how much your data spreads out (variance) along different directions.
2. **Principal Components (PCs):** These are the new axes (eigenvectors) that PCA creates.
3. **Order of Importance:** The first Principal Component PC_1 points in the direction of the most variance, PC_2 in the next most, and so on.
4. **Information Retention:** By selecting the top PCs (those with the largest eigenvalues, representing the most variance), you get a compressed, lower-dimensional view of the data that keeps most of the original information.

How it's Unsupervised:

- **No Labels:** PCA doesn't need a target variable (like "price" or "category") to predict; it works purely on the features (columns) of your dataset.

To verify Coefficients and sorted eigen vectors are matching

Coefficients / coordinates of the PC returned by the 'pca' and the eigen vectors returned by 'eig' may differ in the direction.

Either they exactly alligned in the same direction or they might be pointing at opposite directions.

Sort the PC in the descending order based on the variances.

Take dot product between the corresponding vectors. These vectors are normalized and the only difference between them is in the orientation. So the dot product will be either +1 or -1.

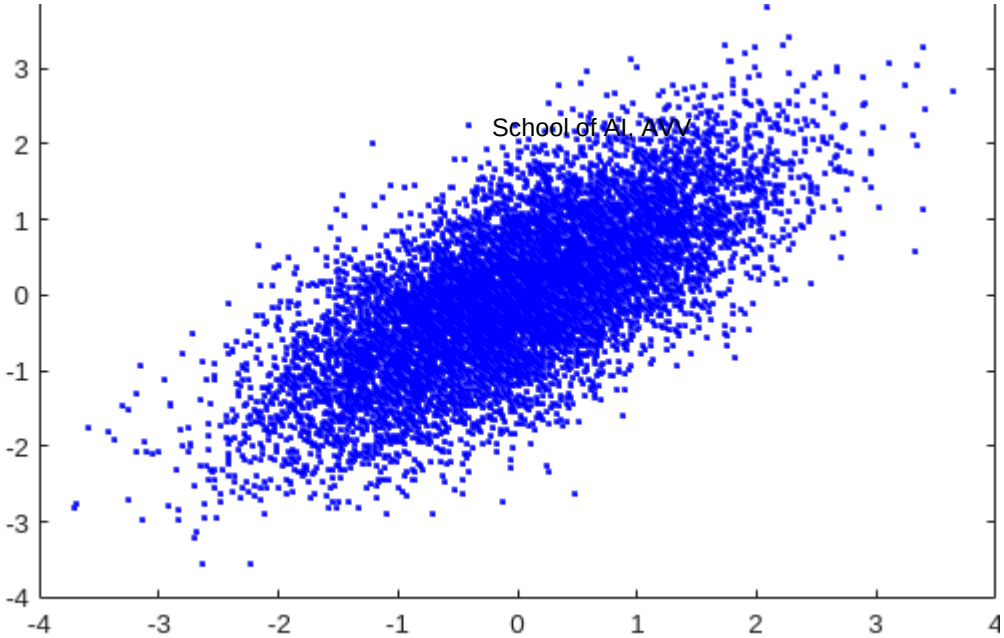
Take the absolute value of the dot product and confirm they are +1.

PCA of correlated data in 2D

```
X = mvnrnd([0 0],[1 0.7; 0.7 1],10000);
size(X)
```

```
ans = 1×2
       10000           2
```

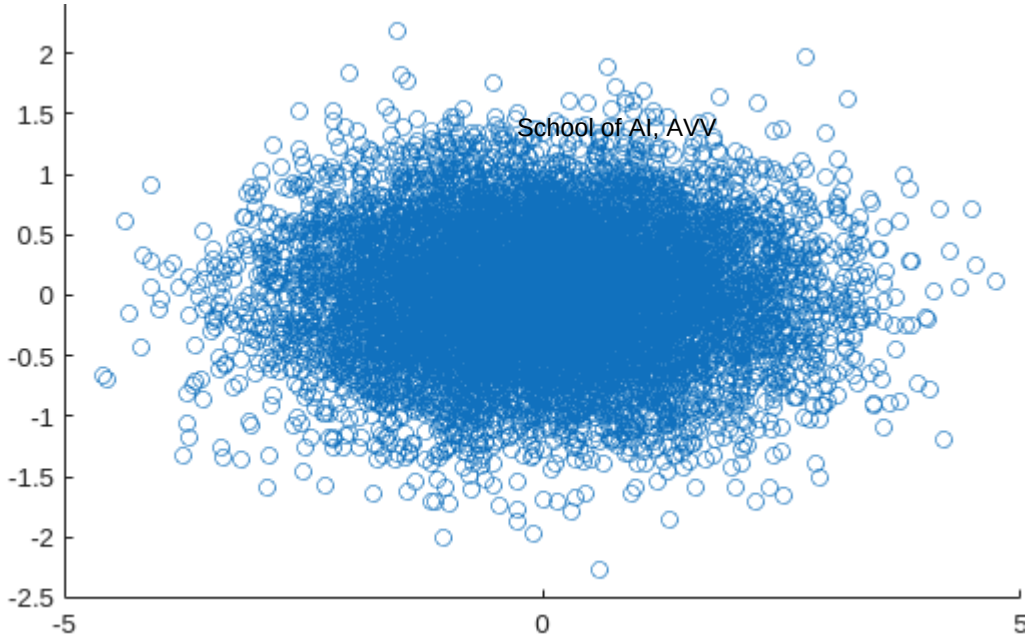
```
plot(X(:,1),X(:,2),'b.')
```




```
[c,s,l] = pca(X)
```

```
c = 2×2
    0.7015    0.7127
    0.7127   -0.7015
s = 10000×2
    2.0416    0.3319
   -0.6289    0.0858
   -0.9279   -0.5844
   -0.5325    0.1980
   -1.4663   -0.7578
   -0.9956   -0.2613
    0.8596   -0.1075
   -2.3478    0.0383
   -0.1050    0.5340
    0.6375   -0.2229
   -2.0130    0.4061
   -0.0958   -0.3470
   -1.6812   -0.2057
    0.4564    0.0989
    2.4200    0.4573
      ⋮
l = 2×1
    1.6943
    0.2966
```

```
scatter(s(:,1),s(:,2));
```




```
% hold on
% scatter(X(:,1),X(:,2),'b.');hold off
```

PCA exercise using iris data set

Lets go step by step with this classic data set. In Matlab, this dataset is available as *"fisheriris.mat".*

```
load fisheriris.mat
whos
```

```
  Name             Size             Bytes  Class      Attributes

  PUBLISH          1x1                  1  logical
  X            10000x2             160000  double
  ans              1x2                 16  double
  c                2x2                 32  double
  l                2x1                 16  double
  meas           150x4               4800  double
  ready            1x1                  1  logical
  s            10000x2             160000  double
  species        150x1              20500  cell
```

The data ia now available in the variable "meas". After loading the dataset it is a good idea to check the variables present in the workspace by

using the command *"whos".* meas has the data and species has the label information. Lets name this as X and

```
X = meas;
```

check its size. We find that there are 150 data points and 4 variables.

```
[Npts,nf] = size(X)
```

```
Npts =
```

```
150
nf =
4
```

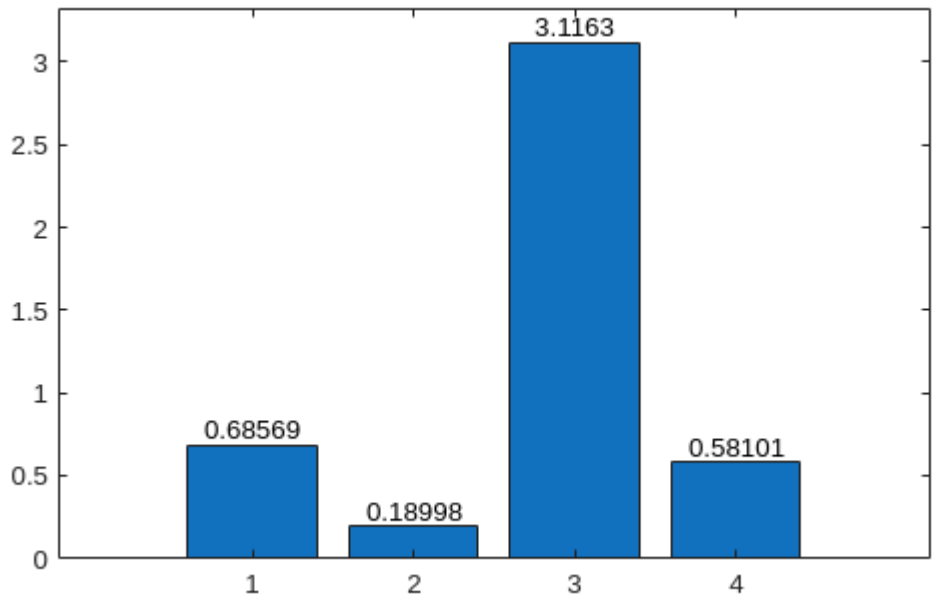
Subtract the mean from the data to centre it around the origin. By doing that we are shifting the entire dataset to the origin and the spread or the variance is not altered.

```
Xo = X - mean(X);
```

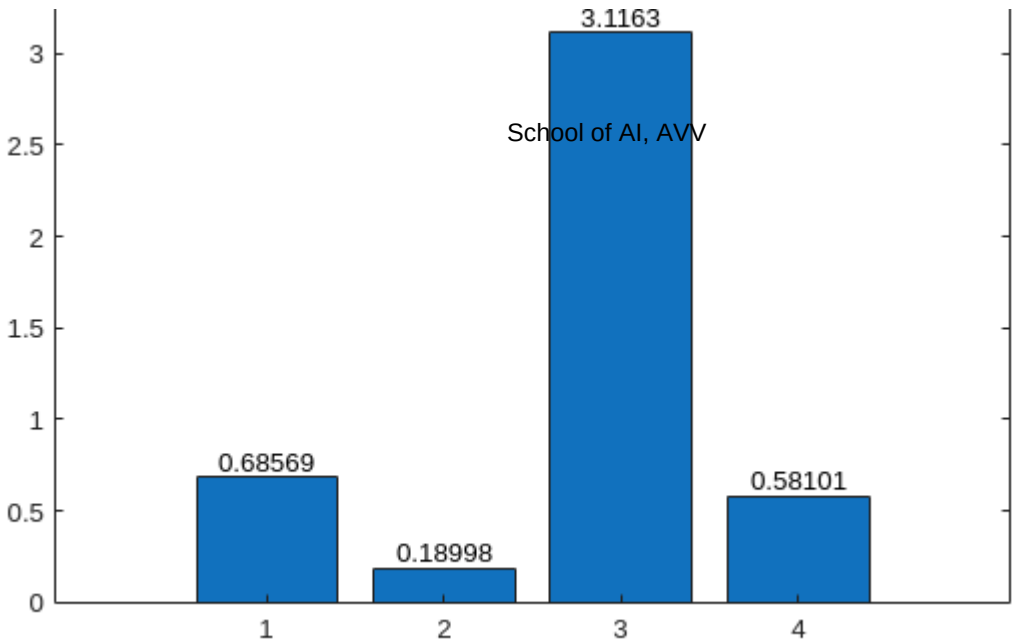
You can check that by computing the variance of the original data and the mean subtracted zero centered data.

Below, bar charts are used to show the variance of the different measurements/observables in the data.

```
b = bar(var(X));
b.Labels = var(X);
```




```
bo = bar(var(Xo));
bo.Labels = var(Xo);
```



Next, we compute the covariance matrix. We can compute from the given data using the formula $\frac{1}{n-1}X_o^T X_o$ or *"use"* the inbuilt function.

```
cv = Xo' * Xo/(Npts-1);
cov(Xo)
```

```
ans = 4×4
    0.6857   -0.0424    1.2743    0.5163
   -0.0424    0.1900   -0.3297   -0.1216
    1.2743   -0.3297    3.1163    1.2956
    0.5163   -0.1216    1.2956    0.5810
```

As expected we get a 4×4 sized covariance matrix. The diagonal elements are the variance of the 4 variables petal width/length and sepal width/length. Since these are the variances of actual measurements they cannot be negative. They can only be zero or positive.

The off diagonal elements in the covariance matrix gives the pair wise covariances of any 2 variables.

cov(x,y) > 0 indicates x and y are positively correlated

cov(x,y) ~ 0 indicates x and y are not correlated

cov(x,y) < 0 indicates x and y are negatively correlated

Lets find the principal components of the data now. These are the directions in which data is spread more. We prefer to orient the axis of measurement along these directions. The principal component directions are given by the eigen vectors of the covariance matrix.

```
[eVec,eVal] = eig(cv);
```

The eigen values give the variance of the data. So the larger the eigen value , the larger the spread. So we sort the eigen values in the descending order and sort the corresponding eigenvectors in the same order.


```
[~,idx] = sort(diag(eVal),"descend");
s_eVec = eVec(:,idx);
```

To project the data along the principal directions, we just multiply the data matrix and the eigen vectors, (since the data is present in the rows, otherwise take the transpose : Important !)

```
Xp = Xo*s_eVec;
```

Now we can take the variance of the data (projected in these directions) !

```
vXp = var(Xp)
```

```
vXp = 1×4
    4.2282    0.2427    0.0782    0.0238
```

Compute the cumulative sum of variances to determine how much information is retained by adding each principal component.

```
cumsum(vXp)*100/sum(vXp)
```

```
ans = 1×4
   92.4619   97.7685   99.4788  100.0000
```

To retain at least 95% information from the raw data, we need to keep 2 principal components.

In other words, we can afford to reduce the dimension of the iris dataset from 4 to 2 and still retain $\approx 97.7\%$ of the total information.

The principal components, the projected data and the variances are returned by the inbuilt function "pca"

as coeff, score and latent respectively.

```
[coeff,score,latent]=pca(X);
```

Eigen values of the covariance matrix (zero centred data) is the

variance of the features.

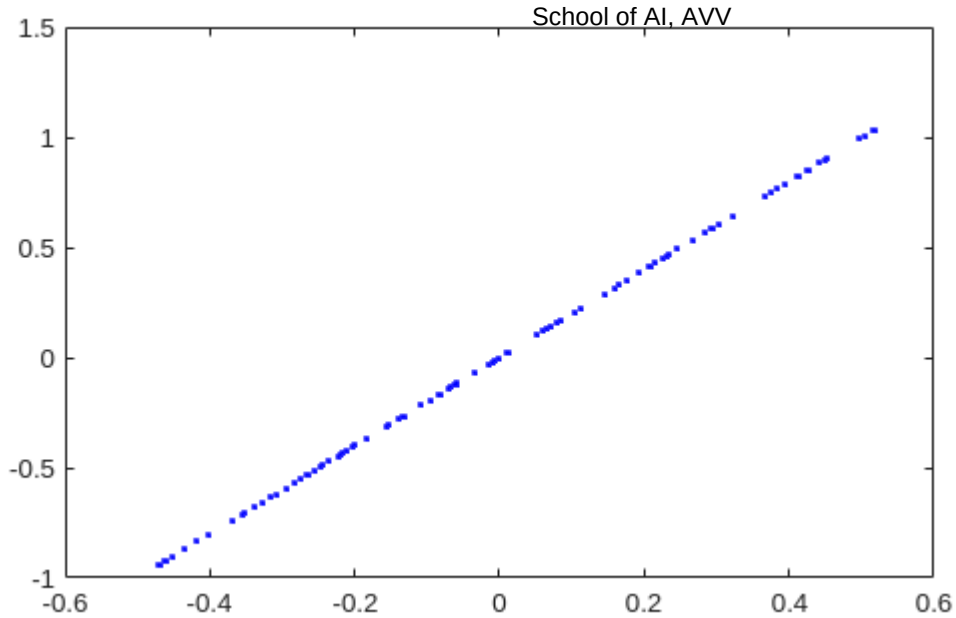
Variance, eigen values or Frobenius norm

PCA of perfectly correlated data in 2D

```
% clearvars

N = 100;
x = rand(N,1);
y = 2*x + 3;% + 0.1*rand(N,1);
X = [x,y];
```

```
X = X-mean(X);
plot(X(:,1),X(:,2),'b.')
```


```
% xlim([0,1]);
% hold off
% ylim([3,5])

cv = X'*X/(N-1)
```

```
cv = 2×2
    0.0807    0.1613
    0.1613    0.3226
```

```
cov(X)
```

```
ans = 2×2
    0.0807    0.1613
    0.1613    0.3226
```

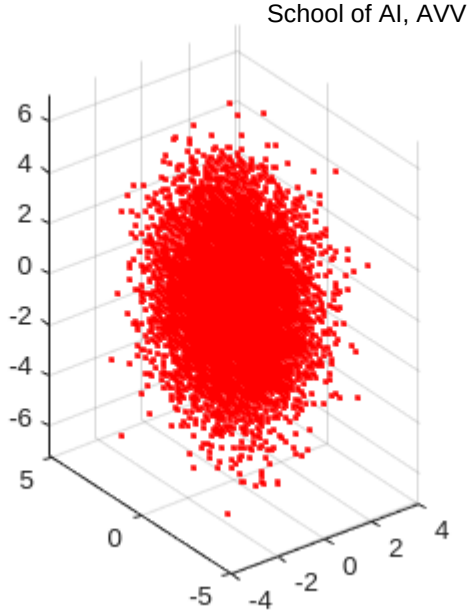
```
% clearvars

N = 1E4;
mu = [4 4 4];
sig = [1 0 0;
       0 2 0;
       0 0 4];
X = mvnrnd(mu,sig,N);
mean(X)
```

```
ans = 1×3
    4.0072    3.9888    4.0172
```

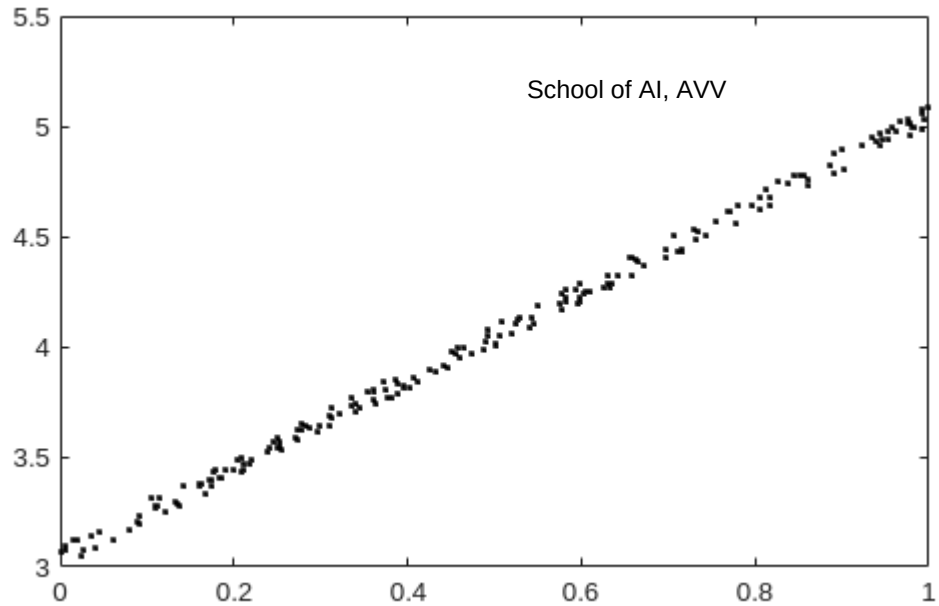
```
Xo = X - mean(X);
```

```
scatter3(Xo(:,1),Xo(:,2),Xo(:,3),'r.')
axis equal
```


```
% clearvars
rng(100)

N = 200;
x = rand(N,1);
y = 2*x + 3  + 0.1*rand(N,1);
X = [x y];
plot(X(:,1),X(:,2),'k.')
```



data for PCA using mvnrnd

```
% clearvars
rng(100)

N = 10;
X = mvnrnd([2,3],[1 0.85; 0.85 0.8],N)
```

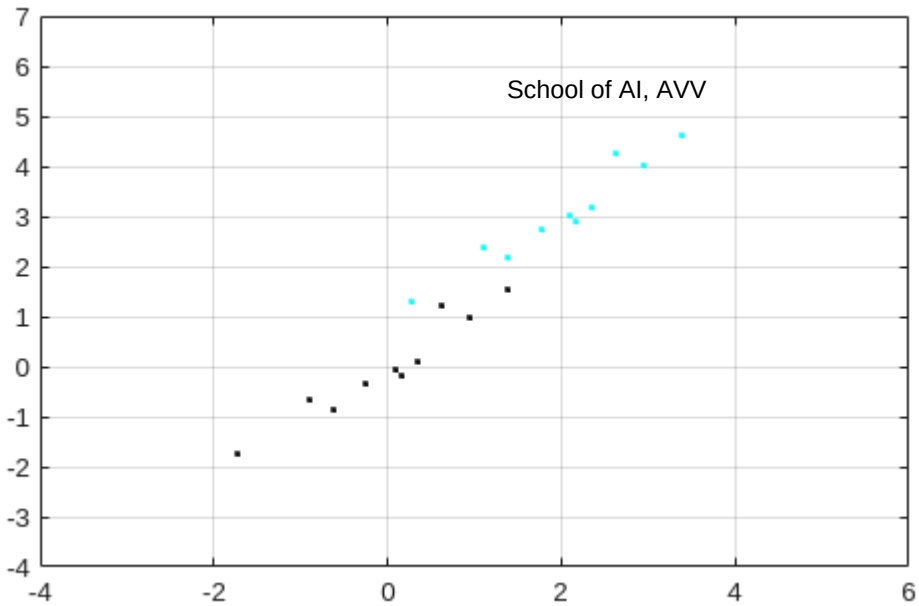
```
X = 10×2
    2.1609    2.8816
    1.3849    2.1888
    1.7610    2.7345
    2.6150    4.2750
    1.0986    2.3786
    2.3481    3.1679
    2.9480    4.0190
    0.2701    1.3061
    2.0971    3.0096
    3.3786    4.6126
```

```
size(X)
```

```
ans = 1×2
    10     2
```

```
plot(X(:,1),X(:,2),'c.');hold on

Xo = X - mean(X);
plot(Xo(:,1),Xo(:,2),'k.');hold off
xlim([-4 6])
ylim([-4 7])
grid on
```



Compute the covariance and compare it with the inbuilt function's (cov) result.

```
cv = Xo'*Xo/(N-1)
```

```
cv = 2×2
    0.8415    0.8958
    0.8958    1.0286
```

```
cv1 = cov(X)
```

```
cv1 = 2×2
    0.8415    0.8958
    0.8958    1.0286
```

Compute the eigen vectors (the principal components) and sort them in descending order.

```
[eVec,eVal] = eig(cv);
[s_eVal,idx] = sort(diag(eVal),"descend");
s_eVec = eVec(:,idx);
```

Project the data on the principal component direction

```
Xp = Xo*s_eVec;
```

Compute the variance

```
var(Xp)
```

```
ans = 1×2
    1.8358    0.0344
```

the principal components, the projected data and the variances are returned by the inbuilt function "pca"

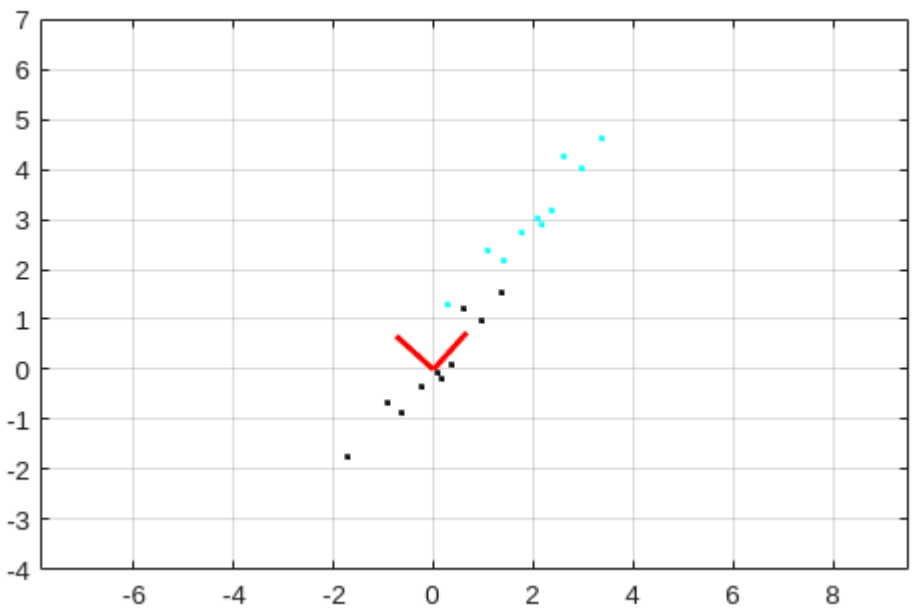
as coeff, score and latent respectively.

```
[coeff,score,latent]=pca(Xp);
latent
```

```
latent = 2×1
    1.8358
    0.0344
```



```
line([0 s_eVec(1,1)],[0 s_eVec(2,1)],LineWidth=2,Color="#FF0000")
line([0 s_eVec(1,2)],[0 s_eVec(2,2)],LineWidth=2,Color="#FF0000")
axis equal
```




```
[coeff,score,latent] = pca(X)
```

```
coeff = 2×2
    0.6694    0.7429
    0.7429   -0.6694
score = 10×2
   -0.0270    0.2326
   -1.0612    0.1198
   -0.4040    0.0340
    1.3121   -0.3628
   -1.1118   -0.2200
    0.3109    0.1800
    1.3448    0.0559
   -2.4631   -0.1175
    0.0253    0.0995
    2.0740   -0.0215
latent = 2×1
    1.8358
    0.0344
```

```
[u,s,v] = svd(X);
```

```
% clearvars
load fisheriris.mat
whos
```



```
  Name           Size            Bytes  Class                                    Attributes

  N              1x1                 8  double
  Npts           1x1                 8  double
  PUBLISH        1x1                 1  logical
  X             10x2               160  double
  Xo            10x2               160  double
  Xp            10x2               160  double
  ans            1x2                16  double
  b              1x1                 8  matlab.graphics.chart.primitive.Bar
  bo             1x1                 8  matlab.graphics.chart.primitive.Bar
  c              2x2                32  double
  coeff          2x2                32  double
  cv             2x2                32  double
  cv1            2x2                32  double
  eVal           2x2                32  double
  eVec           2x2                32  double
  idx            2x1                16  double
  l              2x1                16  double
  latent         2x1                16  double
  meas         150x4              4800  double
  mu             1x3                24  double
  nf             1x1                 8  double
  ready          1x1                 1  logical
  s             10x2               160  double
  s_eVal         2x1                16  double
  s_eVec         2x2                32  double
  score         10x2               160  double
  sig            3x3                72  double
  species      150x1             20500  cell
  u             10x10              800  double
  v              2x2                32  double
  vXp            1x4                32  double
  x            200x1              1600  double
  y            200x1              1600  double
```

```
% clearvars
load examgrades.mat
grades
```

```
grades = 120×5
    65    77    69    75    69
    61    74    70    66    68
    81    80    71    74    79
    88    76    80    88    79
    69    77    74    69    76
    89    93    78    77    80
    55    64    60    50    63
    84    83    80    77    78
    86    75    81    87    79
    84    82    86    92    85
    71    70    73    81    79
    81    88    80    79    83
    84    78    80    74    80
```

```
  81    77    81    83    79
  78    66    90    84    75
   ⋮
```



```
whos
```

```
  Name           Size            Bytes  Class                                    Attributes

  N              1x1                 8  double
  Npts           1x1                 8  double
  PUBLISH        1x1                 1  logical
  X             10x2               160  double
  Xo            10x2               160  double
  Xp            10x2               160  double
  ans            1x2                16  double
  b              1x1                 8  matlab.graphics.chart.primitive.Bar
  bo             1x1                 8  matlab.graphics.chart.primitive.Bar
  c              2x2                32  double
  coeff          2x2                32  double
  cv             2x2                32  double
  cv1            2x2                32  double
  eVal           2x2                32  double
  eVec           2x2                32  double
  grades       120x5              4800  double
  idx            2x1                16  double
  l              2x1                16  double
  latent         2x1                16  double
  meas         150x4              4800  double
  mu             1x3                24  double
  nf             1x1                 8  double
  ready          1x1                 1  logical
  s             10x2               160  double
  s_eVal         2x1                16  double
  s_eVec         2x2                32  double
  score         10x2               160  double
  sig            3x3                72  double
  species      150x1             20500  cell
  u             10x10              800  double
  v              2x2                32  double
  vXp            1x4                32  double
  x            200x1              1600  double
  y            200x1              1600  double
```

```
% clearvars
load cereal.mat
whos
```

```
  Name             Size            Bytes  Class                                    Attributes

  Calories        77x1               616  double
  Carbo           77x1               616  double
  Cups            77x1               616  double
  Fat             77x1               616  double
  Fiber           77x1               616  double
  Mfg             77x1               154  char
  N                1x1                 8  double
  Name            77x1             11520  cell
  Npts             1x1                 8  double
  PUBLISH          1x1                 1  logical
  Potass          77x1               616  double
```

```
  Protein         77x1                616  double
  Shelf           77x1                616  double
  Sodium          77x1                616  double
  Sugars          77x1                616  double
  Type            77x1                616  double
  Variables       15x2               4614  cell
  Vitamins        77x1                616  double
  Weight          77x1                616  double
  X               10x2                160  double
  Xo              10x2                160  double
  Xp              10x2                160  double
  ans              1x2                 16  double
  b                1x1                  8  matlab.graphics.chart.primitive.Bar
  bo               1x1                  8  matlab.graphics.chart.primitive.Bar
  c                2x2                 32  double
  coeff            2x2                 32  double
  cv               2x2                 32  double
  cv1              2x2                 32  double
  eVal             2x2                 32  double
  eVec             2x2                 32  double
  grades         120x5               4800  double
  idx              2x1                 16  double
  l                2x1                 16  double
  latent           2x1                 16  double
  meas           150x4               4800  double
  mu               1x3                 24  double
  nf               1x1                  8  double
  ready            1x1                  1  logical
  s               10x2                160  double
  s_eVal           2x1                 16  double
  s_eVec           2x2                 32  double
  score           10x2                160  double
  sig              3x3                 72  double
  species        150x1              20500  cell
  u               10x10               800  double
  v                2x2                 32  double
  vXp              1x4                 32  double
  x              200x1               1600  double
  y              200x1               1600  double
```



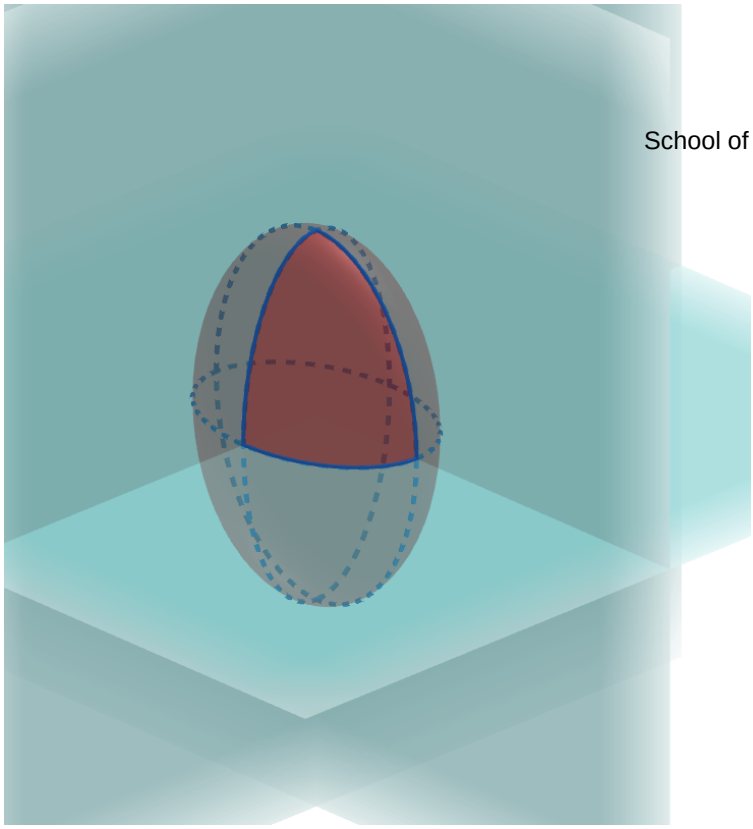
```
Variables
```

```
Variables = 15×2 cell
'Name'      'cereal name'
'Mfg'       'manufacturer (e.g., Kellogg's)'
'Type'      'type of cereal (cold/hot)'
'Calories'  'calories (number)'
'Protein'   'protein(g)'
'Fat'       'fat(g)'
'Sodium'    'sodium(mg)'
'Fiber'     'dietary fiber(g)'
'Carbo'     'complex carbohydrates(g)'
'Sugars'    'sugars(g)'
     ⋮
```

```
% clearvars
load wind.mat
whos
```

Name	Size	Bytes	Class	Attributes
Calories	77x1	616	double	
Carbo	77x1	616	double	
Cups	77x1	616	double	
Fat	77x1	616	double	
Fiber	77x1	616	double	
Mfg	77x1	154	char	
N	1x1	8	double	
Name	77x1	11520	cell	
Npts	1x1	8	double	
PUBLISH	1x1	1	logical	
Potass	77x1	616	double	
Protein	77x1	616	double	
Shelf	77x1	616	double	
Sodium	77x1	616	double	
Sugars	77x1	616	double	
Type	77x1	616	double	
Variables	15x2	4614	cell	
Vitamins	77x1	616	double	
Weight	77x1	616	double	
X	10x2	160	double	
Xo	10x2	160	double	
Xp	10x2	160	double	
ans	1x2	16	double	
b	1x1	8	matlab.graphics.chart.primitive.Bar	
bo	1x1	8	matlab.graphics.chart.primitive.Bar	
c	2x2	32	double	
coeff	2x2	32	double	
cv	2x2	32	double	
cv1	2x2	32	double	
eVal	2x2	32	double	
eVec	2x2	32	double	
grades	120x5	4800	double	
idx	2x1	16	double	
l	2x1	16	double	
latent	2x1	16	double	
meas	150x4	4800	double	
mu	1x3	24	double	
nf	1x1	8	double	
ready	1x1	1	logical	
s	10x2	160	double	
s_eVal	2x1	16	double	
s_eVec	2x2	32	double	
score	10x2	160	double	
sig	3x3	72	double	
species	150x1	20500	cell	
u	35x41x15	172200	double	
v	35x41x15	172200	double	
vXp	1x4	32	double	
w	35x41x15	172200	double	
x	35x41x15	172200	double	
y	35x41x15	172200	double	
z	35x41x15	172200	double	

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$



Matlab datasets

https://in.mathworks.com/help/matlab/import_export/matlab-example-data-sets.html

```
clearvars -except PUBLISH ready
```

```
if(PUBLISH == ready)
    path = '/media/user/DATA4LINUX/new1/Repos/Mine/MFC4_22MAT230/';
    mlxfile = matlab.desktop.editor.getActive().Filename;
    [~, name, ext] = fileparts(mlxfile);
    outfile = [path, name, ext,'.pdf']
    export(matlab.desktop.editor.getActive().Filename, outfile);
    if ispc
        winopen(outfile);
    elseif ismac
        system(['open ' char(outfile)]);
    else
        system("env -u LD_LIBRARY_PATH xdg-open '" + outfile + "' &");
    end
end
```